

Steady-State Electron Transport in Molecular Junctions

Quantum Electronics Final Project

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Introduction to Molecular Junctions

- **The System:** A molecular junction acts as an open quantum system driven far from equilibrium by external leads
- **The Mechanism:** Transport is modeled as a resonant tunneling process bridging discrete molecular energy levels and continuous electrode bands
- **Our Goal:** To numerically simulate and characterize steady-state quantum electron transport through a tight-binding molecular junction driven by a potential bias

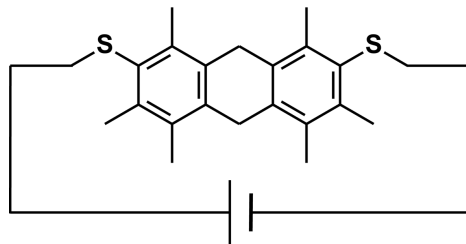


Figure: A schematic molecular junction

Theoretical Framework

- **The Hamiltonian:** The junction is modeled as an open system using a tight-binding Hamiltonian:

$$\hat{H} = \hat{H}_{\text{mol}} + \hat{\Sigma}_L + \hat{\Sigma}_R \quad (1)$$

- **The Green's Operator:** The single-particle scattering dynamics at energy E are solved by inverting the energy matrix:

$$\hat{G}(E) = \lim_{\varepsilon \rightarrow 0} \left[E - \hat{H} + i\varepsilon \right]^{-1} \quad (2)$$

- **Landauer Transport:** The macroscopic steady-state current is obtained using the transmission function $t(E)$ and integrating it across the Fermi window:

$$t(E) = \Gamma_L \Gamma_R |G_{1,M}(E)|^2 \quad (3)$$

$$I = \frac{2e}{h} \int_{-\infty}^{\infty} t(E) [f_L(E) - f_R(E)] dE \quad (4)$$

- The molecule is modeled as a tight-binding uniform linear chain, driven by the following physical parameters:
 - **Chain Length:** A molecule consisting of $M = 3$ sites
 - **On-site Energies:** A non-degenerate configuration with $\alpha = [-0.1, 0, 0.1]$ eV
 - **Internal Hopping:** The electronic coupling strength between adjacent molecular sites is evaluated at $\beta \in \{0.2, 0.5\}$ eV
 - **Electrode's Parameters:**
 - **Potential Bias:** A chemical potential gradient across the electrodes ($\mu_L = 2$ eV, $\mu_R = -2$ eV) under a uniform temperature ($T_L = T_R = 10$ K)
 - **Thermal Bias:** A temperature gradient across the electrodes ($T_L = 30$ K, $T_R = 10$ K) under a uniform chemical potential ($\mu_L = \mu_R = -0.1$ eV)

Potential Bias: Asymmetric Coupling

- Transport saturates as the coupling approaches $2|\beta|$
- Maximum steady-state current is bounded by the weakest electrode interface

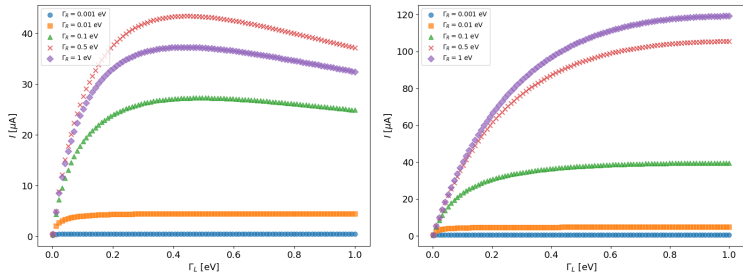


Figure: Current as a function of the coupling strength to the left lead for various fixed coupling strength values to the right lead. Left: $\beta = -0.2$ eV. Right: $\beta = -0.5$ eV.

Potential Bias: Symmetric Coupling

- Resonant transport peaks in the weak-coupling regime before dropping at coupling strengths beyond $\Gamma > 2|\beta|$

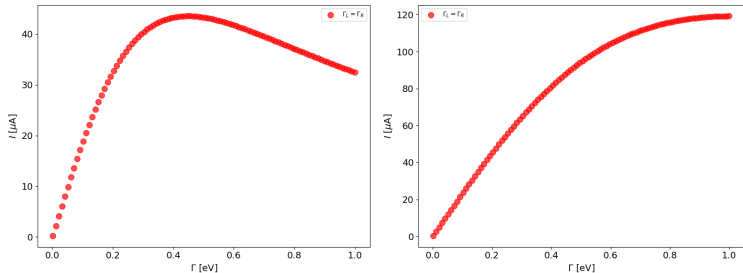


Figure: Current as a function of the coupling strength to both leads. Left: $\beta = -0.2 \text{ eV}$. Right: $\beta = -0.5 \text{ eV}$.

Potential Bias: Global Parameter Landscape

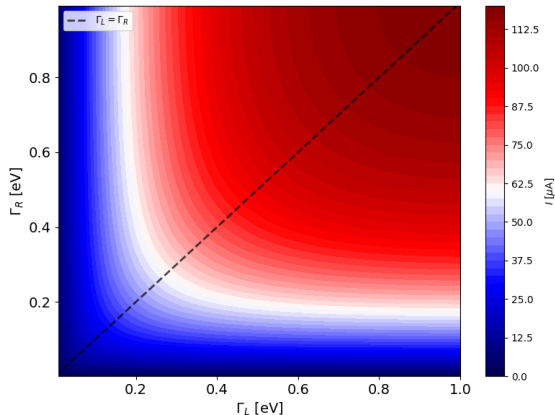


Figure: Current as a function of both lead's coupling strength

Thermal Bias: Symmetric & Asymmetric Coupling

- The peak current is located at the internal molecular hopping parameter $|\beta|$
- When the coupling to the leads is weaker than $|\beta|$ the current increases monotonically with coupling strength.
- when $\Gamma > |\beta|$ the first site hybridizes strongly with the lead, effectively detaching it from the molecule

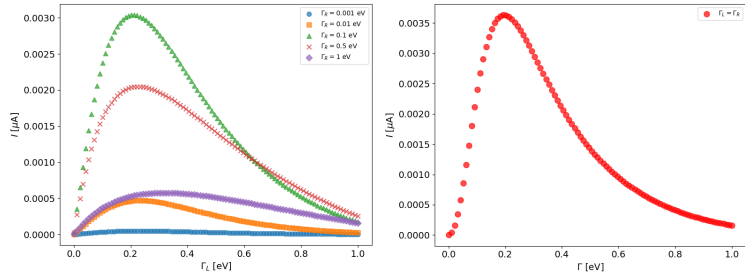


Figure: Current as a function of the coupling strength with $\beta = -0.2$ eV. Left: Asymmetric coupling between the leads. Right: Symmetric coupling between the leads.

Thermal Bias: Global Parameter Landscape

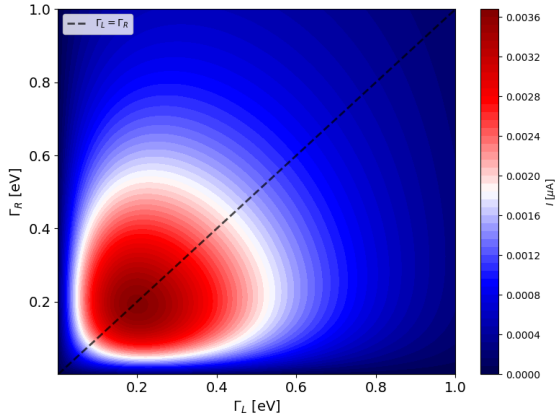


Figure: Current as a function of both lead's coupling strength

- **The Fermi Window:** The current depends heavily on the orbital energy being within the Fermi window. Potential bias adjusts the window width, while thermal bias adjusts the window's edge gradients
- **Coupling Limits:** The maximal current through the system depends on the two hopping rate constants. When the two rates are very different, the current is determined by the slower of the two rates